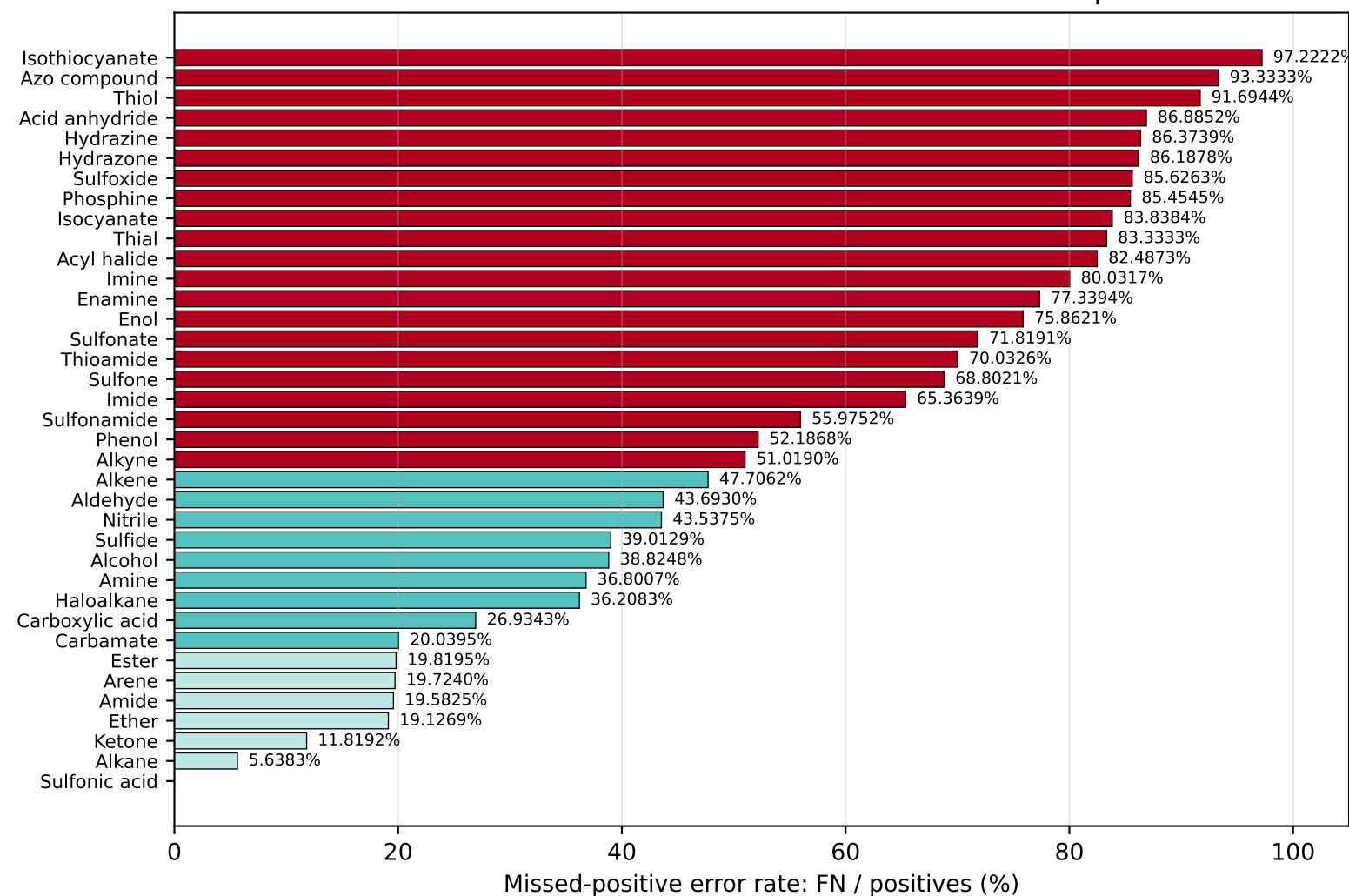
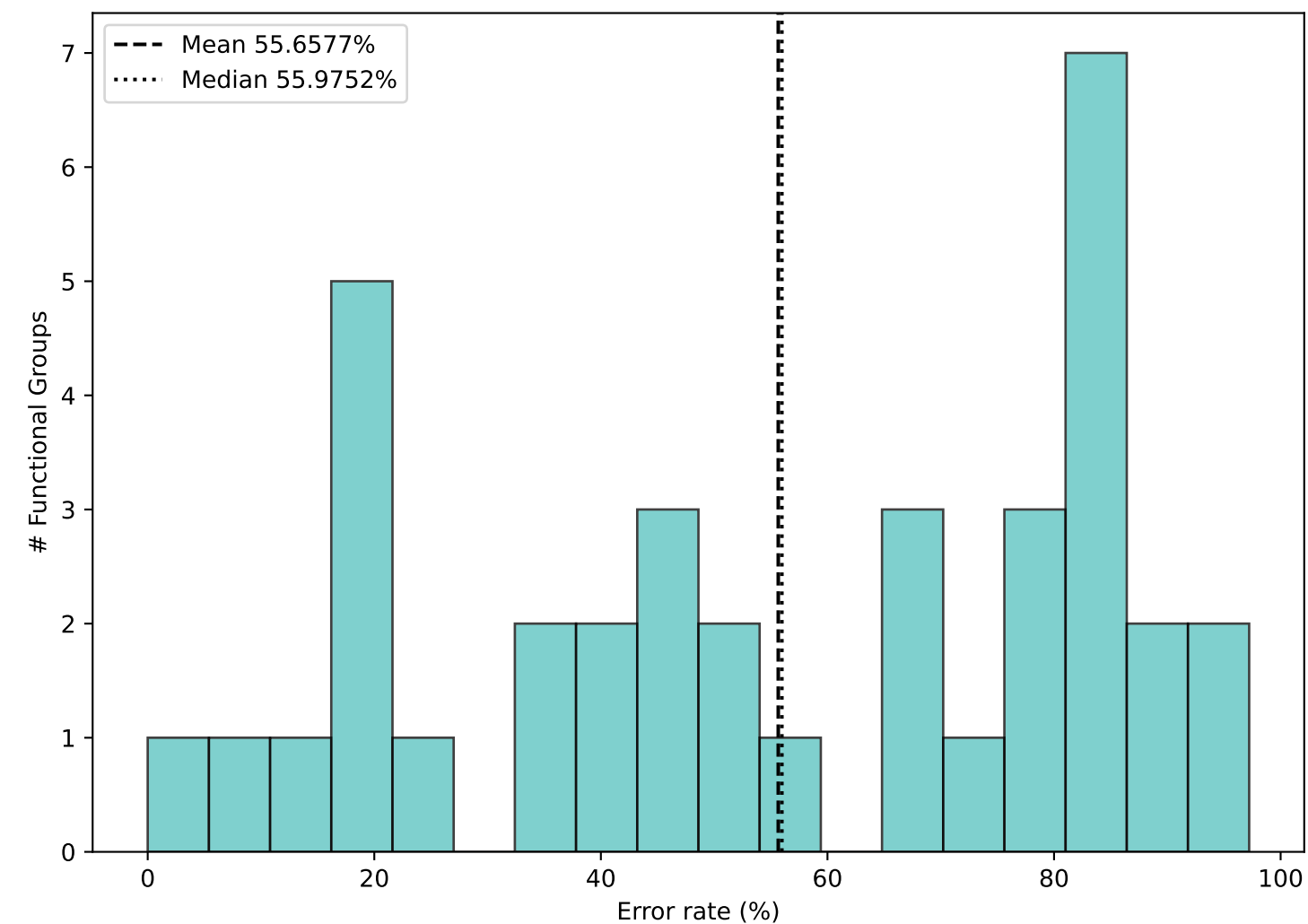


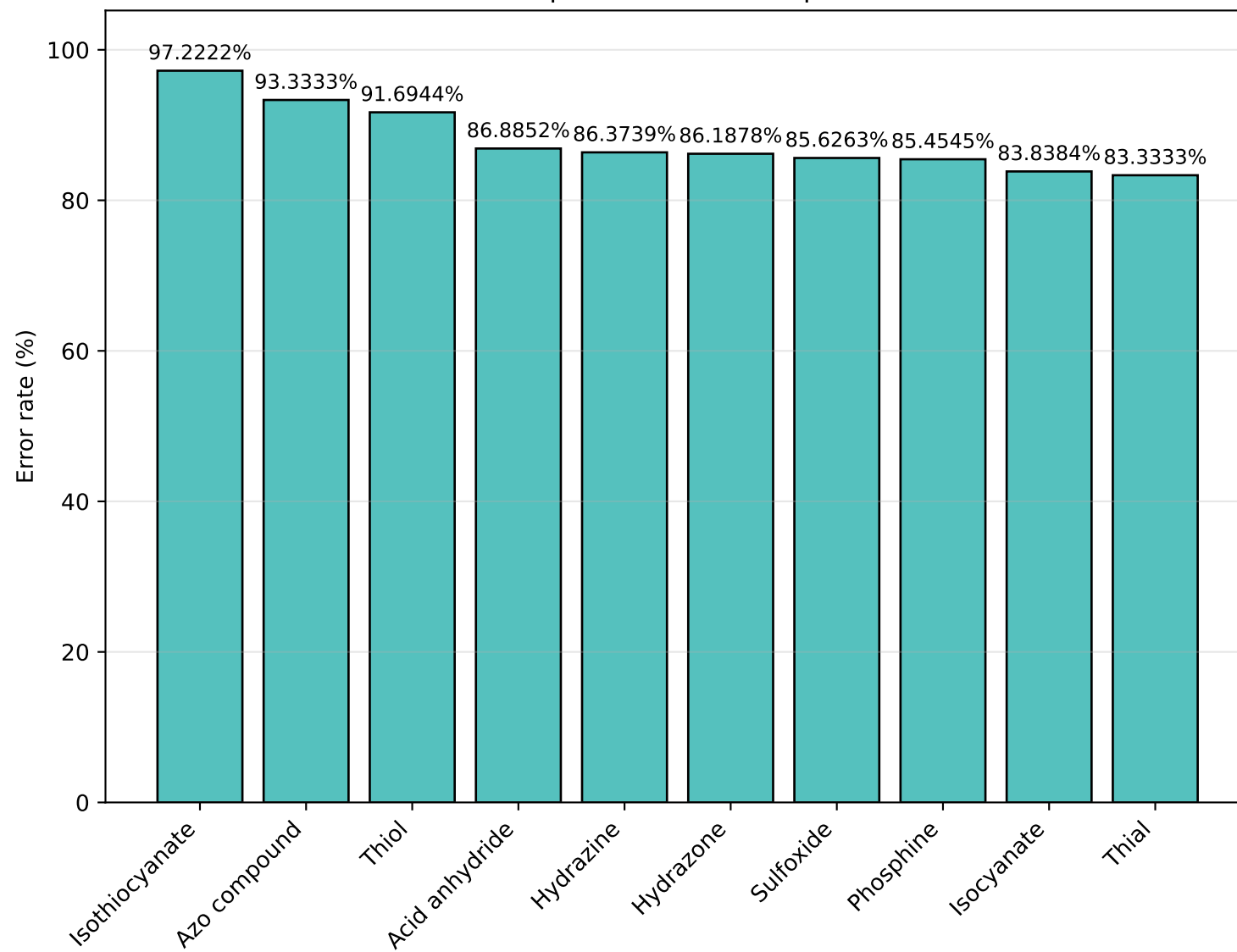
MPS C-NMR Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



MPS C-NMR VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.6470
F1 macro: 0.3324
Precision micro: 0.5689
Recall micro: 0.7499
Hamming accuracy: 0.8943

Error-rate statistics:

Mean: 55.6577%
Median: 55.9752%
Std: 28.9048%
Max: 97.2222% (Isothiocyanate)
Min: 0.0000% (Sulfonic acid)

Problem classes:

Severe (>50%): 21 groups
High (20-50%): 9 groups
Moderate (5-20%): 6 groups
Low (<=5%): 1 groups